Evidence–Rating Alignment for Evidence-Grounded HRMS Performance Evaluation

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Abstract. Employee-performance evaluation in human resource management systems (HRMS) can produce ratings and explanations that appear consistent while lacking criterion-specific evidential support. Agreement between an evaluator and a manager is not, by itself, evidence that the rating is justified. This paper specifies Evidence–Rating Alignment (ERA), a proposed measurement layer for auditing the relationship between role criteria, available evidence, ratings, and explanation claims. ERA combines Evidence Coverage (EC), Evidence Relevance (ER), Criterion Agreement (CA), Evidence Traceability (ET), and a Contradiction Penalty (CP): ERA = 0.25EC + 0.25ER + 0.25CA + 0.15ET + 0.10(1-CP). The paper evaluates the arithmetic and data structure of an existing synthetic pilot workbook containing 100 constructed cases across 10 job roles and 500 criterion-level annotation rows. All 100 reference records remain explicitly provisional, and all core annotation fields are blank; consequently, they cannot serve as independently validated ground truth. Recalculation of the stored ERA composite produced a maximum absolute discrepancy of 0.0006 and no row exceeded a 0.001 tolerance. An exploratory weight-sensitivity analysis on the synthetic component values showed greater rank stability under local perturbations around the proposed weights than under broad perturbations across the weight simplex. These are computational and mathematical audit results only. They do not demonstrate AI accuracy, human agreement, employee-performance validity, fairness, explanation usefulness, or real-world effectiveness. ERA is presented as a transparent, testable design proposal whose validity remains an open empirical question.

Keywords: Explainable AI · Human resource management systems · Employee performance evaluation · Evidence grounding · Auditability · Synthetic data · Composite indicators

1 Introduction

AI-supported HRM can help organize records and support decisions, but employee performance is context-dependent and consequential. Research on AI in

HRM identifies concerns involving data quality, accountability, fairness, privacy, employee experience, and the role of human judgment [1,2,11,12]. Research on explainable AI (XAI) also cautions that an explanation should be evaluated in relation to its purpose and audience rather than assumed trustworthy merely because it is readable [3,4].

Consider a hypothetical employee for whom a manager and an AI system both assign 4/5. The agreement may look reassuring, but both assessments may have omitted a missed deadline or relied on a metric unrelated to the employee's role. Agreement alone cannot reveal whether the rating is supported by appropriate records. This example is illustrative, not an observed organizational case.

This paper addresses the measurement gap between a performance criterion, the evidence available and used, the rating assigned, and the claims offered to justify it. It introduces Evidence–Rating Alignment (ERA) as a proposed audit layer rather than a predictive model or a complete definition of fairness. ERA is intended to make evidence coverage, relevance, reference agreement, traceability, and contradiction visible in a common record. The central contribution is a transparent operational specification and an audit of the currently available synthetic pilot dataset. No new human annotations, real employee records, or fresh LLM outputs are introduced.

2 Research Problem, Objectives, and Contributions

Research problem: how can an HRMS represent whether a criterion-level performance rating is supported by role-relevant evidence, make contradictions and missing support visible, and preserve a traceable record for review and contestation?

The paper has four objectives: (1) define the five ERA components and a normalized composite; (2) specify how the composite can be embedded in an evidence-to-rating workflow; (3) audit the structure and arithmetic of the existing 100-case synthetic workbook; and (4) examine how rankings from its provisional component values respond to alternative weights. These objectives are intentionally narrower than evaluating a live AI system or determining whether ERA improves human decisions.

The contributions are: (a) a criterion-level operational definition of evidence–rating alignment; (b) a proposed audit record linking criteria, evidence identifiers, ratings, contradictions, and review actions; (c) a reproducible dataset and formula audit; and (d) an exploratory weight-sensitivity analysis. The paper does not claim that the composite is validated, that the synthetic reference scores are independent ground truth, or that ERA outperforms existing approaches.

3 Related Work and Research Gap

Responsible-AI research in HRM has examined accountability, privacy, fairness, transparency, and employee involvement [1,2,5,11,12,20,23]. These works motivate governance-aware designs, but broad principles do not themselves specify a

case-by-criterion measure of whether a particular rating is grounded in particular evidence. Reviews of algorithmic decision-making in employment also show that fairness and discrimination concerns arise across recruitment and development settings [5,20].

XAI research offers methods for explaining predictions, including local explanation techniques such as LIME [3,4]. However, explanation fluency or feature attribution is not equivalent to verifying that a record supports a criterion-specific claim. ERA therefore treats evidence relevance and traceability as separate dimensions, rather than treating the presence of an explanation as proof of evidential support.

Research on algorithmic management and organizational justice emphasizes worker experience, power, procedural fairness, dignity, and contestability [6–9,13,14,17,18]. A numerical evidence-alignment score cannot replace these dimensions. A high ERA score should not be interpreted as proof of fairness, legitimacy, or acceptable employment procedure.

A directly related study by Li, Zhou, and Xu examines LLM-based performance evaluation against an external expert-consensus reference [10]. That design illustrates the importance of independent benchmarks when evaluating ratings. The present paper does not run an LLM and has no independently adjudicated benchmark. Its contribution is complementary and methodological: it specifies a candidate evidence-alignment layer and audits the arithmetic of a synthetic prototype. The claimed gap is limited to the selected sources discussed here; it is not a claim that no related method exists anywhere in the literature.

4 Literature Search and Evidence Boundary

The source base is a targeted, structured search for a focused narrative synthesis, not a systematic review. The source manuscript records 11 October 2026 as the search date. Topics included responsible AI in HRM, employee-performance evaluation, explainable AI, algorithmic fairness, human oversight, and benchmark validity. Eligibility focused on English-language scholarly work and authoritative guidance directly relevant to these topics. The search was single-researcher and targeted; independent duplicate screening, a complete deduplication log, formal risk-of-bias assessment, and PRISMA flow counts were not preserved. Therefore, the paper does not claim systematic-review status, comprehensive coverage, or a verified 43-study corpus. Reference [23] is treated only as an external published source.

5 ERA Framework and Mathematical Specification

For case i, let k index predefined role criteria and j index evidence items. Criterion weights w_{ik} are non-negative and sum to one over the criteria assessable for that case. Let r_i be the evaluated rating and b_i the reference rating on the same anchored 1–5 scale. The five ERA components are normalized to [0,1].

A component is undefined if its denominator is zero; in that situation, the case should be flagged and a complete composite should not be silently manufactured.

The proposed composite is:

$$\mathbf{ERA_i = 0.25\ EC_i + 0.25\ ER_i + 0.25\ CA_i + 0.15\ ET_i + 0.10\ (1 - CP_i)}$$

5.1 Evidence Coverage (EC)

Let $A_{ik} = 1$ when criterion k is assessable and 0 otherwise, and $R_{ik} = 1$ when at least one required, relevant evidence item was retrieved for that criterion. Then $EC_i = \Sigma_k\ w_{ik}\ A_{ik}\ R_{ik} / \Sigma_k\ w_{ik}\ A_{ik}$. The denominator must be positive. Coverage measures whether the required evidence was available to the evaluator; it does not establish that the evidence is accurate.

5.2 Evidence Relevance (ER)

For each evidence item actually used, a pre-registered rubric assigns $s_{ij} \in \{1,2,3,4,5\}$; normalize $q_{ij} = (s_{ij} - 1)/4$. With non-negative use weights v_{ij}, $ER_i = \Sigma_j\ v_{ij}\ q_{ij} / \Sigma_j\ v_{ij}$. Equal weights are the default unless another rule is justified before analysis. Evidence that was available but not used is excluded from this component and should be recorded separately as a retrieval omission. If no item was used, ER is undefined.

5.3 Criterion Agreement (CA)

$CA_i = \max(0, 1 - |r_i - b_i|/4)$, where r_i and b_i use the same anchored 1–5 scale. CA expresses agreement with the selected reference, not proof that the reference is correct. Because the current reference values are synthetic and provisional, CA values calculated from them must not be interpreted as agreement with validated employee-performance ground truth.

5.4 Contradiction Penalty (CP)

Let $c_{ij} = 1$ when evidence item j materially contradicts the evaluated claim under a defined coding rubric and 0 otherwise. $CP_i = \Sigma_j\ v_{ij}\ c_{ij} / \Sigma_j\ v_{ij}$ over assessable evidence items used. Missing evidence is not automatically contradictory. If no assessable evidence was used, CP is undefined.

5.5 Evidence Traceability (ET)

Let L_i be the number of atomic rating or explanation claims that require support and T_i the number that include a valid evidence identifier and a verifiable claim-to-record mapping. $ET_i = T_i/L_i$. A citation or identifier alone is insufficient: the record must exist, be accessible to an authorized reviewer, and support the relevant claim. If $L_i = 0$, ET is undefined.

5.6 Weighting and Interpretation

The weights sum to one and are a transparent, theory-led starting configuration, not empirically estimated parameters. EC, ER, and CA each receive 0.25; ET receives 0.15; and CP contributes a 0.10 penalty term. These choices express design priorities rather than proven relative importance. Composite-indicator research recommends explicit weight assumptions and sensitivity checks [21,22]. The additive score is compensatory, so it must be shown alongside component values and flags. A high score must not override missing evidence, material contradiction, or the need for human review, and it must not automatically determine pay, promotion, termination, or other employment outcomes.

6 Proposed XAI–HRMS Workflow and Audit Record

The proposed workflow begins with authorized records and a frozen role-specific criterion rubric. A retrieval or evidence-selection stage links each assessable criterion to evidence identifiers, provenance, source type, and time window. A rating stage records the criterion-level score and the evidence actually used. An audit stage computes the ERA components, records missing and contradictory evidence, and identifies unsupported claims. A human-review stage permits the authorized reviewer to inspect the records, override or abstain from a rating, record a rationale, and initiate correction or appeal. The architecture is a design specification, not a deployed HRMS service.

For each criterion-level output, the audit record should include: case and criterion IDs; rubric version; rating and scale; evidence IDs used; evidence provenance and timestamps; evidence relevance and validity coding; missing or contradictory evidence flags; atomic explanation claims and their evidence links; component values; composite ERA; undefined-component flags; reviewer action and rationale; and a versioned audit timestamp. The record supports traceability and contestability but does not itself prove that a rating is fair or valid.

7 Explanation, Fairness, and Human Oversight Measures

The current dataset contains no generated natural-language explanations that were independently reviewed. Therefore explanation quality is specified, not measured. For a future evaluation, explanations should be decomposed into atomic factual claims. Let N be the number of eligible claims, S the number fully supported, U the number unsupported or contradicted, V the number unverifiable because required records are unavailable, and P the number partially supported. These categories should be mutually exclusive and sum to N. Evidence Support Rate is S/N; Unsupported Claim Rate is U/N; Unverifiable Claim Rate is V/N; and Partial Support Rate is P/N. If N=0, these metrics are undefined.

Evidence identification precision and recall can be calculated against a separately frozen evidence key: EIP = $|R \cap G|/|R|$ and EIR = $|R \cap G|/|G|$, where R is the set of returned evidence IDs and G is the set of required IDs. Their

harmonic mean is reported only when defined. For planted contradictions, error-detection precision, recall, F1, and false-positive rate can be measured. These metrics are not reported as results here because no qualifying model outputs or independently scored explanations are present.

Explanation usefulness should be measured through separately reported reviewer ratings of clarity, evidence sufficiency, actionability, and contestability. Fairness requires a separate analysis of decision processes and error patterns; ERA itself contains no protected-attribute or subgroup fairness term. Sensitive attributes should be used only where lawfully authorized and ethically approved. Meaningful human oversight requires the ability to inspect evidence, challenge a rating, correct records, abstain, and appeal. These are design requirements, not evaluated outcomes of the current study.

8 Data and Methods

8.1 Available Dataset

The analysis uses the existing workbook ERA_Pilot_Dataset_100_Cases_QC_Corrected.xlsx, sheet "100_Cases_QC_Corrected", plus its role-criteria and annotation-form sheets. The case table contains 100 synthetic records, 100 unique case IDs, and 10 roles with 10 cases per role. The designed case-pattern distribution is: aligned (20), ai_human_agree_wrong (20), contradiction (15), strong (15), weak (15), mixed (10), and missing (5). These proportions are features of the constructed dataset and must not be interpreted as the prevalence of these patterns in real organizations.

The workbook also contains 500 criterion-level annotation rows (five per case). The core fields for evidence relevance, evidence validity, criterion benchmark rating, and contradiction coding are blank in all 500 rows. Every case's benchmark status explicitly states that human annotation/adjudication is required before the reference can be treated as validated ground truth. Despite a field name containing "independent benchmark", the scores are rule-generated synthetic anchors; the label does not make them empirically independent or human-validated.

8.2 Analysis Plan

Three analyses were conducted using existing workbook values only. First, a structural audit checked row counts, unique identifiers, role coverage, pattern counts, annotation completeness, and benchmark status. Second, the stored composite was independently recomputed from the documented formula and the workbook's EC, ER, CA, CP, and ET columns; absolute differences were summarized. Third, an exploratory sensitivity analysis recomputed scores under three named alternative weight configurations and 10,000 random weight draws under each of two Dirichlet schemes. The random generator seed was fixed at 20261027. Rank stability was measured with Spearman's rank correlation relative to the proposed weights.

The sensitivity analysis is a mathematical analysis of provisional synthetic component values, not a test of predictive accuracy or construct validity. The source files available for this revision did not include a complete transformation script and per-case output table sufficient to reproduce the earlier described three-condition experiment. Therefore, the earlier condition means and associated rating-error results are not reported in this version. No new LLM/API calls, human annotations, or real employee records were used.

9 Results

9.1 Dataset and Formula Audit

The structural audit identified 100 cases with 100 unique IDs, no duplicate case IDs, and 10 roles represented by 10 cases each. All 100 benchmark records remained provisional. The annotation workbook contained 500 criterion-level rows, with zero rows containing any of the four core rating/coding fields. The stored ERA composite had a mean of 0.80874 and ranged from 0.667 to 0.947; these are descriptive summaries of the supplied synthetic records, not model performance results.

Audit item	**Result**	**Interpretation**
Synthetic cases / unique IDs	100 / 100	No duplicate case IDs found.
Roles	10; 10 cases per role	Balanced by construction.
Criterion annotation rows	500	Five criterion rows per case.
Rows with any core annotation field completed	0 of 500	No human annotations are present.
Provisional benchmark status	100 of 100 cases	Reference scores are not validated ground truth.
Mean absolute difference: stored vs recomputed ERA	0.000265	Consistent with rounding in the workbook.
Maximum absolute difference	0.000600	No row exceeded the 0.001 tolerance.
Rows with difference $>$ 0.001	0	Arithmetic consistency check passed at the stated tolerance.

The formula audit supports a narrow conclusion: the stored composite values are arithmetically consistent with the documented equation to within the specified tolerance. It does not validate the component annotations, synthetic reference scores, weights, or substantive meaning of the score.

9.2 Exploratory Weight Sensitivity

Under the proposed weights, the mean composite recalculated from the available component fields was 0.808754. Three named alternative weight sets produced Spearman rank correlations of 0.9446 (equal weights), 0.9685 (evidence-heavy), and 0.9552 (traceability-heavy) relative to the proposed weights. In the 10,000-draw local Dirichlet analysis centered on the proposed weights, the median rank correlation was 0.9639 (5th–95th percentile: 0.8844–0.9917), and 92.13% of draws had $\rho \geq 0.90$. Under broad Dirichlet(1,1,1,1,1) perturbations, the median was 0.7542 (5th–95th percentile: 0.4768–0.9342), and 11.10% of draws had $\rho \geq 0.90$.

Weight scheme	**Mean composite**	**Spearman ρ vs proposed**
Proposed: 0.25, 0.25, 0.25, 0.15, 0.10	0.808754	1.0000
Equal: 0.20 each	0.805490	0.9446
Evidence-heavy: 0.30, 0.30, 0.15, 0.15, 0.10	0.801874	0.9685
Traceability-heavy: 0.20, 0.20, 0.20, 0.25, 0.15	0.805065	0.9552

Random-weight scheme	**Draws**	**5th percentile ρ**	**Median ρ**	**95th percentile ρ**	**Share with $\rho \geq 0.90$**
Local: Dirichlet(40 × proposed weights)	10,000	0.8844	0.9639	0.9917	92.13%
Broad: Dirichlet(1,1,1,1,1)	10,000	0.4768	0.7542	0.9342	11.10%

The sensitivity analysis indicates that rankings of these synthetic cases are more stable when weights remain near the proposed configuration and less stable when weights vary broadly. These proportions describe analyst-selected mathematical perturbations; they are not probabilities that the weights are correct. Because the underlying component values are synthetic and provisional, the analysis does not establish optimal weights, real-world rank stability, or validity of employee ratings.

10 Discussion

10.1 What the Current Analysis Establishes

The available evidence supports three limited conclusions. First, the ERA equation is explicitly defined and can be recalculated from the supplied normalized component fields. Second, the workbook has a balanced synthetic structure across ten named roles, while its 500 annotation rows remain unfilled and its reference status remains provisional. Third, case rankings based on these synthetic components are relatively stable under local weight perturbations but more sensitive under broad perturbations. These conclusions are reproducible from the included workbook and script.

10.2 What It Does Not Establish

The analysis does not show that ERA identifies correct employee-performance ratings, that an AI system generated the listed illustrative ratings, that a human reviewer agrees with the benchmark, or that ERA improves decisions. The workbook's "illustrative_ai_rating" field is not accompanied by a model identifier, prompt, run log, or raw model output; it is therefore treated as an illustrative synthetic field rather than a reproducible AI result. The synthetic reference scores are not independent human ground truth. No human inter-rater reliability, explanation faithfulness, fairness, reviewer usefulness, or organizational deployment result is available.

10.3 Implications for HRMS Design

ERA is best understood as a candidate audit layer that makes evidence coverage, relevance, agreement, contradiction, and traceability explicit. In an HRMS, component-level visibility may help a reviewer identify why a composite score is high or low, but the score should never conceal an undefined component or a critical contradiction. A system should display the component profile and the underlying records, allow abstention when evidence is insufficient, and preserve a contest path. These are design implications, not claims that such benefits have been demonstrated.

10.4 Research Questions Revisited

RQ1 (operationalization) is addressed at the specification and arithmetic-audit level. RQ2 (response to controlled evidence changes) is not claimed as a finding in this version because the available condition-level transformation artifacts were insufficient to reproduce the earlier reported experiment. RQ3 (incremental value over simpler baselines) remains unanswered because no independent outcome is available. RQ4 (weight sensitivity) is addressed only as a mathematical sensitivity analysis on synthetic components. RQ5 (reliability, usefulness, fairness, and external validity) remains outside the evidence currently available. These boundaries are part of the result, not a request that the current paper be supplemented with invented data.

11 Validity, Ethics, and Limitations

Construct validity is not established: the component definitions are a proposal and have not been validated by a representative expert panel or against independent outcomes. Criterion validity is not established because the reference scores are synthetic and provisional. Internal validity for an evidence-intervention effect cannot be assessed without reproducible condition transformations and outputs. External validity is not established because no real employee records or organization were studied. Computational reproducibility is supported for the formula audit and weight sensitivity analysis; human annotation reliability was not measured.

The dataset is synthetic, with 100 constructed cases across 10 roles. Its pattern frequencies are designed and do not estimate real-world prevalence. All 500 core annotation records are blank. The workbook includes rating-like strings that were capped at the stated 5-point scale in a quality-corrected copy; the correction log preserves the edits and raw evidence strings. This correction improves scale consistency but does not validate the content. The illustrative rating field lacks documented model provenance. Weight sensitivity depends on analyst-selected perturbation distributions and provisional inputs.

No real employee data, personally identifiable information, human participant data, or external LLM/API calls were used in the current computational audit. This statement does not assert that institutional ethics approval or exemption has been granted. Any later study involving human participants or real employee data must follow the relevant institutional and legal review, data minimization, access control, retention, and governance requirements. ERA must not automatically determine employment outcomes; meaningful review and contestability are necessary safeguards.

12 Conclusion

This paper presents Evidence–Rating Alignment (ERA) as a proposed measurement layer for auditing whether employee-performance ratings are supported by criterion-relevant evidence. It defines five normalized components—coverage, relevance, criterion agreement, traceability, and contradiction penalty—and specifies how they can be embedded in an auditable HRMS workflow. An audit of the available 100-case synthetic workbook found 100 unique cases across 10 roles, 500 uncompleted criterion-level annotation rows, provisional benchmark status for every case, and stored composite values that matched the documented equation within a maximum absolute discrepancy of 0.0006. An exploratory sensitivity analysis showed stronger rank stability under local than broad weight perturbations on these synthetic values.

The contribution is a framework specification and reproducible computational audit, not a validated employee-performance instrument. No human or LLM results are claimed, and no real-world accuracy, fairness, or decision-improvement effect is inferred. The paper is complete within that declared scope: it reports